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The impact of perceived partisanship on climate policy support: A conceptual replication and extension of the temporal framing effect

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ABSTRACT

Bridging the political divide between liberals and conservatives is one of the biggest challenges in reaching broad public support of climate policies. Research has suggested that framing climate policies with respect to the past may reduce opposition by political conservatives, but recent attempts to replicate this effect have failed. A new perspective on these inconsistent findings may be offered by taking the influence of temporal framing on individuals' perception of the messenger into account. The present work investigated how implicit cues contained in temporal message framing as well as explicit political identity cues shape the perceived political orientation of a messenger and subsequent climate policy support by partisans. Across three experiments ($N_{total} = 2012$), we found that past (vs. future) framing and conservative (vs. liberal) party affiliation independently contributed to the messenger being perceived as more conservative. Past framing and conservative party affiliation increased endorsement of the messenger and the message by conservatives, but decreased endorsement by liberals. However, past framing and conservative party affiliation independently increased conservatives' climate policy support, with mixed effects on liberals. Moreover, a temporal framing effect only emerged when messenger party affiliation was made explicit, suggesting that activating individuals' political identity facilitates the integration of implicit identity cues contained in temporal framings. We discuss the theoretical implications of our integrated account for the observation of partisan effects and reassess the potential of temporal framings to reduce the political divide on climate change.

1. Introduction

Research from the political sciences, economics, and psychology provides abundant evidence for the predictive power of citizens' political orientation on their attitudes, beliefs, and actions [Jost, 2006; Hornsey et al., 2018]. Political conservatism has been identified as a major driver of climate change scepticism [Hornsey et al., 2016; Hornsey & Fielding, 2020] and a barrier to climate policy support [Goldberg et al., 2020; Marghetis et al., 2019]. Over the last decades, the polarization between liberals and conservatives seems to have even increased, posing a serious barrier to climate change mitigation [Dunlap et al., 2016; Hornsey et al., 2018]. However, the policy preferences of liberals and conservatives have also shown to be malleable, suggesting that partisan preferences are at least partly constructed in the context of a given situation rather than reflections of deeply rooted values and goals [Lichtenstein & Slovic, 2006; Vosgerau & Peer, 2019; Zaller, 1992; Cohen, 2003]. The presence of policy-unrelated information such as proof of one's social group or the language used to promote a given policy may thus importantly influence partisans' context-dependent policy preferences [Baldwin et al., 2018; Cohen, 2003].

Emerging research suggests that conventional message framing, i.e., advocating for climate policies in ways that mostly appeal to the liberal side of the political spectrum, may drive and exacerbate the political divide [Lammers & Baldwin, 2018; Graham et al., 2009; Baldwin & Lammers, 2016; Feinberg & Willer, 2013]. The lower support of conservatives for climate policies may partly stem from these differences in message framing rather than from fundamental differences in conviction about how to react to climate change [Van Boven et al., 2018].

Building on these insights, research has started to investigate message framings that systematically vary contextual information to reduce the political divide in climate policy support [Hardisty et al., 2010; Petrovic et al., 2014; Feinberg & Willer, 2019].

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In the present research, we revisit research on temporal message framing [i.e., past vs. future framing; Baldwin and Lammers, 2016; Lammers & Baldwin, 2018] which found past framings to selectively increase conservatives' environmentalism. However, these original findings were not replicated in several recent attempts [Stanley et al., 2021; Kim et al., 2021]. By highlighting the commonalities of temporal message framing and explicit messenger identity labeling (e.g., party affiliation) in influencing individuals' perception of the messenger, we offer a new perspective that may help to explain these inconsistent findings.

In the following, we will review the literature on framing interventions aiming to increase conservatives' environmentalism, the existing literature on temporal framing effects, and the literature on the influence of explicit messenger identity on individuals' judgments and decisions [i.e., partisan effects Cohen, 2003; Unsworth & Fielding, 2014; Diamond & Zhou, 2021; Iyengar et al., 2019], concluding with our proposed conceptual framework of the influence of implicit temporal and explicit political identity cues on individuals' perception and judgments.

1.1. The impact of framing on conservatives' climate policy support

Framing - the highlighting of specific aspects of information through formats that trigger different psychological processes - is a powerful behavior change tool [Kahneman & Tversky, 1984; Tversky & Kahneman, 1981; Beshears & Kosowsky, 2020]. Framing emphasises a specific aspect of information without significantly changing its content, such as highlighting the potential *benefits* as opposed to *risks* of introducing a new drug [Tversky & Kahneman, 1981]. Accordingly, framing constitutes a subtle and low-cost behavior change strategy that operates without monetary incentives nor restrictions to the decision makers' freedom of choice [Thaler & Sunstein, 2008]. In the context of the political divide in climate policy support, research has identified a number of framings that were found to increase the support of conservatives for climate policies.

For instance, research has investigated whether single words that are traditionally associated with a conservative rather than a liberal political agenda may increase conservatives' climate policy support [Hardisty et al., 2010; Schuldt et al., 2011; Petrovic et al., 2014]. In line with this idea, conservatives more strongly supported a policy offsetting the emissions from flying when the policy was described as *carbon offset* instead of a *carbon tax* [Hardisty et al., 2010]. Similarly, self-identified Republicans more readily accepted that climate change is real when referred to as *climate change* rather than *global warming* [Schuldt et al., 2011].

Moreover, highlighting how environmental protection serves the more conservative moral values of loyalty, authority, or sanctity (i.e., moral framings) was found to increase conservatives' climate change beliefs, conservation intentions, and support for a U.S. transition away from fossil fuels toward cleaner energies [Wolsko et al., 2016; Wolsko, 2017; Feinberg & Willer, 2019; Feinberg & Willer, 2013; Hurst & Stern, 2020].

1.2. Temporal framing effects and need for additional research

Research has suggested that *temporal framings* are another promising strategy of increasing conservatives' support of traditionally more liberal (climate) policies. Throughout a series of studies, conservatives more strongly endorsed liberal political goals as diverse as leniency in justice, gun control, immigration, social diversity, social justice, and climate change mitigation when the political goals were framed with references to the past rather than to the future [Lammers & Baldwin, 2018; Baldwin & Lammers, 2016]. For example, one study showed that conservatives reported higher attitudes towards protecting the environment after reading a message advocating to combat climate change to reestablish the past, as compared to a message advocating to combat climate change to build a better future [Baldwin & Lammers, 2016]. Conversely, this research found that the higher attitudes of liberals towards liberal goals, such as environmental protection, remained generally unaffected by different temporal framings [Lammers & Baldwin, 2018; Baldwin & Lammers, 2016].

In light of the potential theoretical and practical relevance of temporal framing effects, several studies have attempted to replicate and extend previous research, with mixed results [Kim et al., 2021; Stanley et al., 2021]. In two high-powered direct replications of Study 1 from Baldwin and Lammers (2016), Kim et al. (2021) did not replicate that a past-framing leads to higher environmental attitudes of conservatives in comparison to a future-framing. Although the authors did observe an interaction of temporal framing and political orientation for the endorsement of an environmental message, a more fine-grained analysis indicated that, contrary to the original findings, endorsement was lower for liberals exposed to the past-framed (vs. future-framed) message, while the endorsement of conservatives did not vary as a function of temporal framing [Kim et al., 2021].

Stanley et al. (2021) found similar results when aiming to replicate Study 3 from Baldwin and Lammers (2016). Although the same materials were used and additional relevant environmentalism variables were measured (e.g., willingness to make lifestyle changes and climate policy support), the interaction of political orientation and temporal framing was not replicated for any of the environmental outcome variables except for *climate change certainty* [Stanley et al., 2021]. However, post-hoc analyses revealed that differences in climate change certainty were only partially driven by the expected increase of certainty among conservatives, but also by a decrease of certainty among liberals [Stanley et al., 2021].

In summary, recent failures to replicate the temporal framing effect have raised doubts about its robustness, practical relevance, and nature [Stanley et al., 2021; Kim et al., 2021]. Specifically, these findings suggest that a past framing does not systematically promote environmentalism among conservatives and that it may have a negative effect on the environmentalism of liberals. Importantly, the back-firing effect on liberals' environmentalism cannot be accounted for by the original theoretical model, which exclusively predicts a past framing to increase the environmentalism of conservatives (although a similar back-firing effect was observed, but not discussed, for liberals' environmental donations in Study 6 of Baldwin and Lammers (2016).

To reconcile these inconsistent findings, the present work proposes a novel theoretical perspective that integrates the temporal framing effect with the literature on partisan effects [Cohen, 2003; Iyengar et al., 2019]. Specifically, we propose that temporal framings provide implicit cues about the political identity of the messenger, with past framings signaling that a messenger is rather conservative and future framings signaling that a messenger is rather liberal. If a past framing leads to a messenger being perceived as more conservative, the previously observed decreases in messenger endorsement, certainty about climate change, and environmental donations by liberals may be explained by a weaker perceived alignment of liberal participants' own political orientation with the orientation of the messenger perceived as being conservative [Kim et al., 2021; Stanley et al., 2021; Baldwin & Lammers, 2016].

Conceptualizing temporal message framing as an implicit cue of messenger identity draws a parallel to research investigating the influence of explicit political identity cues (e.g., party affiliation) on policy support. In what follows, we review this literature and propose an integrated research approach investigating both implicit and explicit messenger identity cues.

1.3. Effects of partisan cues on policy support

The influence of political messenger identity on policy support has most prominently been investigated as the *party-over-policy* effect, which describes the phenomenon that liberals' as well as conservatives' policy support is more strongly determined by the position of their respective political party than by their own evaluation of the policy. In a seminal series of experiments, self-identified liberals and conservatives expressed their attitudes toward a welfare policy program that was either supported by U.S. Democrats or Republicans [Cohen, 2003]. The results showed that information on the approval of a given policy by the Democratic or Republican party almost exclusively determined partisans' policy support: Liberals supported the welfare policy when it was supported by the Democratic Party (but not by the Republican Party) and conservatives supported the identical policy when it was supported by the Republican Party (but not by the Democratic Party). Strikingly, this effect was largely independent of the welfare policy being more generous - thus fundamentally aligning with the values of liberals, or more stringent - fundamentally aligning with the values of conservatives (see also Bullock (2011)). In another study, self-identified Republicans and Democrats were found to report more favorable attitudes and stronger policy support when a carbon tax and a nuclear energy policy were endorsed by members of their respective political party [Fielding et al., 2020]. Extensive research has provided supporting evidence for the party-over-policy effect [Guilbeault et al., 2018; Van Boven et al., 2018; Hart & Nisbet, 2012; Iyengar et al., 2019; Fernbach & Van Boven, 2022; Westfall et al., 2015].

1.4. Integration of implicit and explicit political identity cues

Most research has investigated the alignment of individuals' policy support with the position of political parties by providing explicit party- or identity cues (i.e., Democrat/Republican or liberal/conservative). We want to argue here that a temporal framing contains more subtle, implicit identity cues which individuals use in addition to explicit cues to make inferences about the political identity of a messenger. Liberals have a preference for future-oriented social change and progress while conservatives tend to prefer past-oriented tradition and stability [Jost et al., 2009]. These preferences are prominently reflected by the temporal references made by liberal and conservative political elites [Lammers & Baldwin, 2018]. For instance, a content analysis of 145 State of the Union addresses revealed that Republican presidents more frequently mentioned the past than the future, while the opposite was true for Democratic presidents [Robinson et al., 2015]. Individuals are likely to have learned this association between political affiliation and the use of temporal references and may thus perceive a messenger using future message framing to be more liberal and a messenger using past message framing to be more conservative. Consequently, *implicit* (i.e., temporal framing) cues of the political identity of a messenger may add to the influence of *explicit* (i.e., party affiliation) cues in shaping individuals' perception of the political orientation of a messenger. In turn, the alignment of perceived messenger orientation and individuals' own political orientation can be expected to produce a combined, indirect partisan effect on judgments and policy support.

Previous research on cue utilization supports the interplay of implicit and explicit identity cues in judgment and impression formation [Brunswik, 1956; Slovic, 1966; Anderson, 1962]. According to Brunswik's lens model [Brunswik, 1956], individuals utilize available cues to form their judgments of an unobservable criterion (e.g., the political identity of a person). Moreover, when integrating multiple cues to form their judgments, individuals tend to linearly integrate available cues, especially in the presence of few cues [Karelaia & Hogarth, 2008] and when they consistently point into the same direction of a judgment [Slovic, 1966; Anderson, 1962]. Research in the context of hiring decisions supports the usefulness of a lens-model approach to investigate the integration of implicit as well as explicit identity cues into judgments [Derous & Decoster, 2017; Kleissner & Jahn, 2021]. For example, implicit (e.g., candidate name) and explicit (e.g., candidate age) cues resulted in similar discrimination of older job candidates in a hiring scenario presented to recruitment professionals [Kleissner & Jahn, 2021]. In line with accounts of judgment and impression formation, the influence of implicit and explicit age cues on hiring decisions was additive, meaning that none of the two cues canceled out the influence of the other.

Although previous research on temporal framing did experimentally vary the explicit identity of a messenger to be either a self-identifying liberal or conservative, it did not investigate the influence of explicit identity cues beyond testing for a potential moderation of the temporal framing effect, which was not observed [Kim et al., 2021; Baldwin & Lammers, 2016]. To the best of our knowledge, no previous research has systematically investigated the integrated effects of implicit temporal and explicit political identity cues on individuals' perception of the messenger and subsequent judgments and environmental outcomes.

1.5. Aim of the present research

The present research aims to provide a novel perspective on temporal framing effects by theoretically linking it to research on partisan effects [Cohen, 2003; Iyengar et al., 2019]. We propose that - in addition to explicit political identity cues such as party affiliation - temporal message framings implicitly influence the perceived political orientation of the messenger (see Fig. 1). Perceived political orientation of the messenger, in turn, is expected to influence messenger endorsement and environmental outcomes along the entire spectrum of individuals' own political orientation. This perspective offers an explanation for findings of liberals being equally affected by temporal framings under some circumstances [Stanley et al., 2021; Kim et al., 2021; Baldwin & Lammers, 2016] because it predicts that temporal framings influence conservatives as well as liberals through their perception of the messenger. Finally, by varying the affiliation of the messenger to either a liberal or conservative political party the present research also aims to inform the practical question of who might effectively use temporal framings to increase conservatives' climate policy support.

1.6. Overview and hypotheses

We address the research questions developed above in the context of policy support for electric vehicle subsidies in Switzerland, where electric vehicles hold an important potential to decarbonize the transportation sector [SFOE, 2017]. Climate policy making shows considerable political polarization among Swiss citizens [Hahnel et al., 2020], although this political polarization is likely to be less pronounced than in the U.S. [Smith & Mayer, 2019; Czarnek et al., 2021; Hornsey et al., 2018]. These circumstances provide a conservative test of temporal framing effects in a novel geographical and political context. Another particularity of the Swiss context is that its political system allows for direct participation of citizens in the legislative process through public referendums. As a consequence, voting for policy implementation constitutes a regular part of Swiss citizens' duties; a circumstance that increases the ecological validity of the present research.

Experiment 1 investigated the influence of implicit identity cues contained in temporally framed messages (i.e., past vs. future) on individuals' perception of the political orientation of the messenger (a-path in Fig. 1), tested for the occurrence of a temporal framing effect (c-path), explored an indirect partisan effect constituted by the interaction of perceived political orientation with individuals' own political orientation (d-path). Experiment 2 held temporal framing constant by merely using a future-framed message, while investigating the influence of different explicit political identity cues (i.e., liberal vs. conservative party affiliation) on individuals' perception (b-path) and environmental outcomes (e-path). Experiment 3 investigated the combined influence of implicit temporal identity cues and explicit political identity cues on individuals' perception of the political orientation of the messenger (a- and b-path), individuals' endorsement of the message and its messenger, and their climate policy support (c- and e-path).

Hypothesis 1. (Experiment 1): We expected that a past-framed

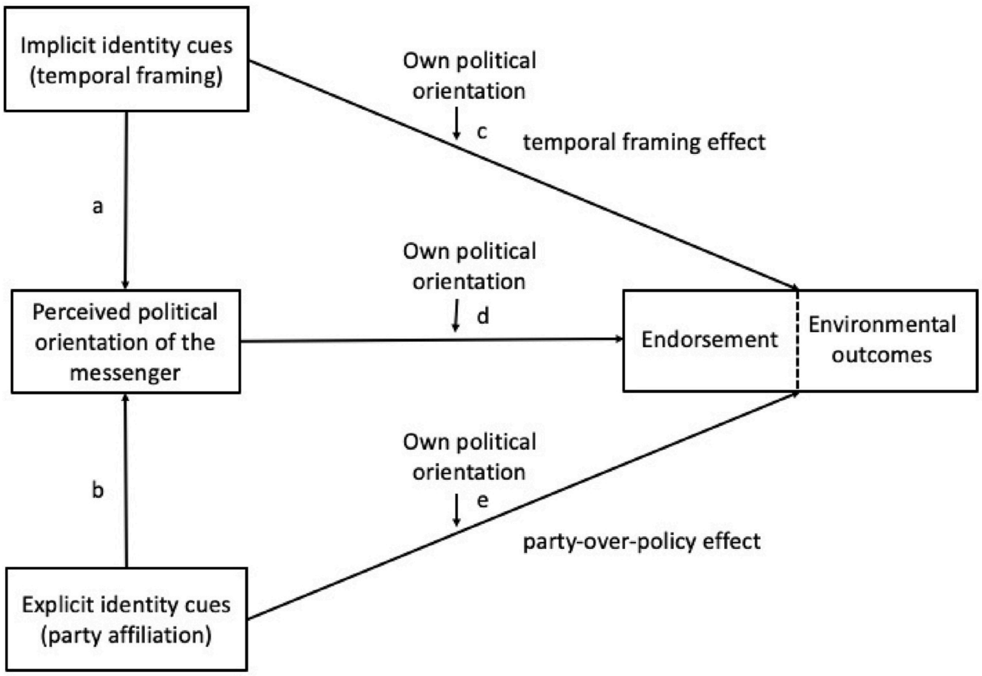

Fig. 1. The proposed conceptual framework of the relationship between implicit political identity cues (temporal framing = future/past), explicit political identity cues (political party affiliation = liberal/conservative), the perceived political orientation of the messenger, individuals' own political orientation, the endorsement of the messenger/an environmental message, and other environmental outcomes such as climate policy support. The letters *c* and *e* indicate the interactions predicted by the temporal framing effect [Baldwin & Lammers, 2016] and the party-over-policy effect [Cohen, 2003]. Table 1 in the Appendix summarizes the direct and indirect effects observed throughout the three experiments of the present research.

message, by containing implicit political identity cues, leads individuals to perceive a messenger to be more politically conservative than a future-framed message (a-path in Fig. 1).

Hypothesis 2. (Experiment 2): We expected that an explicit political identity cue (i.e., the party affiliation of the messenger to a conservative [liberal] party), increases conservatives' [liberals'] and decreases liberals' [conservatives'] endorsement and climate policy support (e-path in Fig. 1).

Hypothesis 3. (Experiment 3): We expected that implicit (i.e., temporal framing) and explicit (i.e., party affiliation) identity cues independently contribute to the perception of the political orientation of the messenger (a-path and b-path in Fig. 1).

Given the repeated failures to replicate the temporal framing effect in previous research, we did not explicitly formulate a hypothesis about the observation of a temporal framing effect on endorsement and climate policy support in Experiment 1 and 3 (c-path in Fig. 1). However, based on our account and previous findings we considered it likely that we would not only observe an effect of temporal framing for conservatives [Baldwin & Lammers, 2016; Lammers & Baldwin, 2018], but also for liberals [Stanley et al., 2021; Kim et al., 2021]. We left it open to exploration if potential variations in outcomes along the political spectrum reflect an attenuation of the political divide or rather the alignment of partisans with their (perceived) in-group [Cohen, 2003; Iyengar et al., 2019].

Moreover, we explored to what extent the temporal framing and party-over-policy effect (c- and e-path) could be accounted for by the mediating role of the perceived political orientation of the messenger (d-path).

2. Experiment 1

Experiment 1 was designed to test the influence of temporal framing on individuals' perception of the political orientation of the messenger (Hypothesis 1; a-path in Fig. 1). Additionally, Experiment 1 constituted a conceptual replication of the original temporal framing effect [Baldwin & Lammers, 2016] with individuals' endorsement of the environmental message and its messenger, and climate policy support as outcomes. We tested the temporal framing effect using two different versions of a newspaper article advocating for public subsidies of electric mobility in Switzerland.

2.1. Participants

A sample of Swiss citizens (N = 507, 260 women) was recruited online via a market research institute and provided informed consent to their participation in accordance with the requirements of the local ethics commission. Sample sizes were in line with previous replication research [Kim et al., 2021], ensuring a statistical power of $1 - \beta > 0.80$ at a critical p value of .05 (two-tailed test) to detect the original temporal framing effect of $b = 0.42$ [Study 1 in Baldwin & Lammers, 2016]. In order to increase the ecological validity of our findings, all participants were required to have the Swiss nationality and to be eligible to vote. The mean age was 49.2 years (SD = 13.98). Distributions of sex, age, and political orientation closely represented the general Swiss population (see Table 4 of the Appendix). The completion of the experiment took about 5 min for which participants were compensated with 2€.

2.2. Procedure and experimental design

After providing demographic information, participants reported their political orientation on a scale ranging from 1 = *extremely left* to 10 = *extremely right*, conforming with the labeling of liberal and conservative political orientation commonly used in a European context. In the remainder of the article, we will use *liberal* when referring to a left-wing political orientation and *conservative* when referring to a right-wing political orientation [Jost, 2006]. Participants' political orientation was normally distributed and showed sufficient variation to meaningfully investigate this variable in our sample (M = 5.3, SD = 1.8). Participants read a short introduction stating that electric mobility is a currently much-discussed topic in politics and industry. They were additionally informed that public subsidies to support the technology were controversial and that they would read a newspaper article which positions itself with regard to public subsidies for electric mobility in Switzerland. Next, depending on the experimental condition, participants were presented with either a future- (n = 265) or past-framed (n = 242) article of about 200 words encouraging Swiss citizens to vote for electric vehicle subsidies in a public referendum (see the Appendix for the complete stimuli). Next, participants answered two questions with respect to the content of the article to assure they had carefully read it. Participants who wrongly answered at least one of the questions were not allowed to continue with the survey ($n = 34$) and were replaced by newly recruited participants to maintain the planned sample size of 500. Then participants reported their voting intentions for the discussed subsidies in a hypothetical public referendum on a scale from 1 = *absolutely opposed* to 6 = *absolutely in favor*. Participants were then asked to evaluate how they perceived the political orientation of the messenger on a scale from 1 = *extremely left* to 10 = *extremely right*. To stay in line with the terminology used in previous temporal framing research (Baldwin et al., 2016; Kim et al., 2021; Stanley et al., 2021), the present work refers to *conservative* for a right-wing political orientation and to *liberal* for a left-wing political orientation.[1] completed the following three measures adapted from previous research and representing a mix of messenger and message endorsement [Lammers & Baldwin, 2018]: To what extent do you agree with the content of the article?; To what extent do you perceive the author of the article as likeable?; Would you vote for the author of the article if s/he ran for office?; all measured on a scale from 1 = *not at all* to 5 = *absolutely yes*. Cronbach's alpha of the endorsement measures yielded high consistency ($\alpha = 0.89$, 95%*CI*[0.87, 0.91]) which allowed us to combine the measures into an endorsement index [Lammers & Baldwin, 2018]. Across all participants, support of the policy ($M = 3.95$, $SD = 1.44$) and level of endorsement ($M = 3.09$, $SD = 0.88$) was moderate. All measures obtained in this and the following experiments, except for participants' party vote during the last federal elections, are included in our analyses and reported in the manuscript. Table 4 in the Appendix reports the bivariate correlations between all variables included in Experiment 1.

2.3. Results and discussion

First, we tested for a temporal framing effect by evaluating the impact of the temporal framing on individuals' endorsement of the messenger and the message, and climate policy support (c-path in Fig. 1). We computed separate linear regressions including participants' self-reported political orientation, temporal framing, and the interaction of the two as predictors of endorsement and policy support. ANOVA results did not show the interaction of participants' political orientation and temporal framing as predicted by the original account [Baldwin & Lammers, 2016; Lammers & Baldwin, 2018], neither for endorsement nor policy support ($F(1, 503) = 0.06$, $p = .807$ and $F(1, 503) = 0.21$, $p = .647$). Inspection of the main effects of political orientation and temporal framing revealed that conservatives reported lower endorsement and policy support than liberals ($t(503) = -3.17$, $b = -0.17$, $se = 0.06$, $p = .002$ and $t(503) = -4.05$, $b = -0.36$, $se = 0.09$, $p < .001$), and that a past-framed message, unexpectedly, increased overall endorsement and policy support independent of participants' political orientation ($t(503) = 3.77$, $b = 0.29$, $se = 0.08$, $p < .001$ and $t(503) = 2.02$, $b = 0.25$, $se = 0.12$, $p = .044$). While this overall positive effect across the political spectrum may have been driven by the novelty of a past framing in the context of the advocacy for electric vehicles subsidies, the non-replication of an interaction between temporal framing and participants' political orientation was in line with previous unsuccessful replications [Kim et al., 2021; Stanley et al., 2021].

Next, we tested the influence of temporal framing on the perception of the political orientation of the messenger (Hypothesis 1; a-path in Fig. 1). In line with electric mobility being rather a political goal of liberals in Switzerland, perception of the political orientation of the messenger was somewhat skewed towards the liberal extreme ($M = 4.4$, $SD = 1.7$), but was still sufficiently normally distributed with skewness $= 0.41$ and kurtosis $= 3.25$, staying within the recommended boundaries for normal distribution of skewness $< |2|$ and kurtosis $< |7|$ [Hair et al., 2010]. In line with Hypothesis 1, the results of a linear regression revealed that the messenger was perceived as more conservative when communicating a past-framed message than when communicating a future-framed message ($t(505) = 2.35$, $b = 0.35$, $se = 0.15$, $p = .019$; see Fig. 2). This result provided first evidence for our account and suggested that individuals derive conclusions about the political identity of the messenger from the temporal framing of a message.

To further examine the role of participant's perception of the political orientation of the messenger, we then analyzed whether messenger perception indirectly influenced endorsement and policy support (d-path in Fig. 1). The results of two linear regressions revealed a statistically significant interaction of perceived political orientation and participants' own political orientation for endorsement and policy support ($F(1, 503) = 44.32$, $p < .001$ and $F(1, 503) = 25.07$, $p < .001$; see Fig. 3). In line with the party-over-policy effect [e.g., Cohen, 2003], the regression coefficients indicated that the more strongly participants' own political orientation overlapped with their perceived political orientation of the messenger, the higher their endorsement and policy support ($t(503) = 6.66$, $b = 0.07$, $se = 0.01$, $p < .001$ and $t(503) = 5.01$, $b = 0.09$, $se = 0.02$, $p < .001$). Given that temporal framing altered the perception of messenger identity (a-path), this suggests an indirect framing effect with perceived political orientation of the messenger mediating the relationship between temporal framing and (i) individuals' endorsement of the messenger and (ii) their climate policy support (upper triangle in Fig. 1).

To probe for this indirect effect, we computed the "PROCESS" macro model 15, for R v4.2, with bias-corrected 95% confidence intervals ($n = 10{,}000$) and assessed the statistical significance of the moderated mediation index (i.e., MM_{index}) by examining whether the bootstrap confidence intervals of the conditional indirect effect excluded zero [Hayes, 2017]. The results showed that perceived political orientation of the messenger, moderated by individuals' own political orientation, mediated the relationship between temporal framing and endorsement ($MM_{index} = 0.025$, 95%*CI*[0.004, 0.048]) as well as policy support ($MM_{index} = 0.031$, 95%*CI*[0.005, 0.061]; see Table 1 in the Appendix for detailed results of the mediation analysis). Given that we could not exclude the possibility of unobserved variables confounding the relationship between messenger perception and endorsement/policy support based on our data [Bullock & Green, 2021; Rohrer et al., 2022], this finding should be considered exploratory and interpreted with caution.

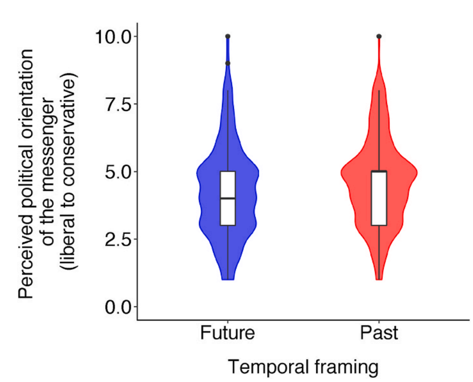

Fig. 2. Perception of messenger political orientation in Experiment 1. Box- and density plots illustrating the influence of temporal framing on the perceived political orientation of the messenger. A future framing resulted in a more liberal perception of the political orientation of the messenger, while a past framing resulted in a more conservative perception of the political orientation of the messenger. Perceived political orientation was reported on a scale from 1 = *extremely left* to 10 = *extremely right*. Points show outliers that are situated more than 1.5 times the interquartile range away from the upper limit of the box plot.

[1] We discuss the (non-)equivalence of *right* = *conservative* and *left* = *liberal* in the general discussion.

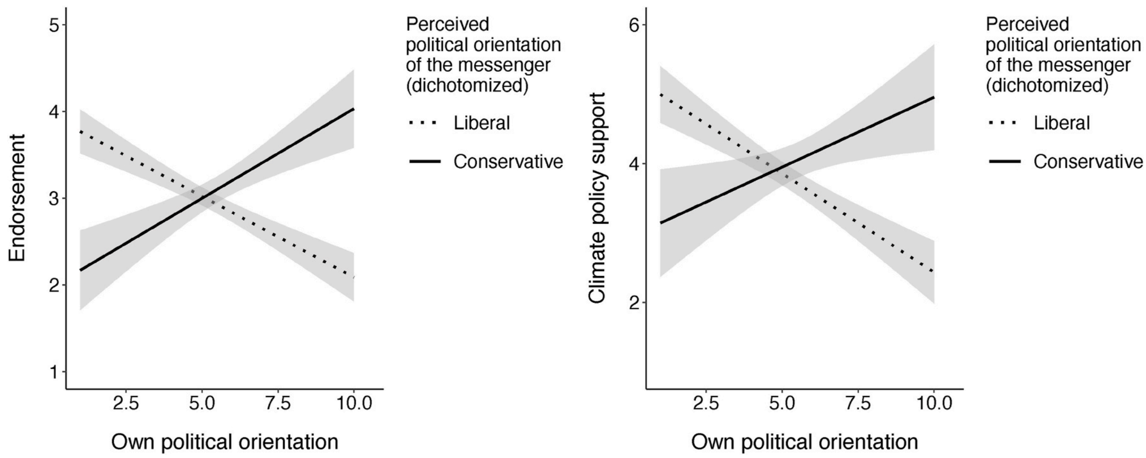

Fig. 3. Endorsement and climate policy support in Experiment 1. The interaction of participants' overall perception of the political orientation of the messenger, dichotomized for illustrative purposes to > 5: Conservative and ≤ 5: Liberal, with their own political orientation on endorsement and policy support. Endorsement was an aggregate of three measures, ranging from 1 = *low endorsement* to 5 = *high endorsement*, constituting a mix of messenger and message endorsement [see Methods for details; Lammers & Baldwin, 2018]. Climate policy support was measured on a scale from 1 = *absolutely opposed* to 6 = *absolutely in favor*. Own and perceived political orientation was reported on a scale from 1 = *extremely left* to 10 = *extremely right*. Grey areas represent 95% confidence intervals around the regression lines.

In summary, Experiment 1 did not replicate a direct effect of temporal framing on conservatives' endorsement and policy support [Baldwin & Lammers, 2016]. In line with our assumptions, however, we found supporting evidence for the influence of temporal framing on the perception of the political orientation of the messenger (a-path in Fig. 1) and for the importance of perceived political orientation in shaping individuals' endorsement and policy support (d-path). Mediation analysis supported perceived political orientation of the messenger as a mediator of the effect of temporal framing on endorsement and policy support. These results suggest that perception of messenger identity plays a central role in shaping individuals' environmental judgments and policy preferences. Experiment 2 followed up on this finding by experimentally testing whether a less ambiguous, explicit cue of political messenger identity similarly shapes messenger perception, endorsement, and climate policy support.

3. Experiment 2

Experiment 2 tested the hypothesis that an explicit political identity cue leads to increased endorsement of the messenger and the message as well as increased climate policy support, depending on individuals' own political orientation (Hypothesis 2; e-path in Fig. 1). To this end, we explicitly manipulated messenger identity by labelling the author of a future-framed article to be either affiliated to a conservative or a liberal Swiss political party. We merely used the future framing of the article since electric vehicles are more commonly discussed in a future oriented way, providing an adequate context to study the influence of messenger party affiliation. In contrast to the implicit temporal identity cues used in Experiment 1, we expected that an explicit identity cue would less ambiguously influence participants' perception of the political orientation of the messenger, resulting in an interaction of messenger party affiliation with participants' own political orientation [Cohen, 2003]. Importantly, this test of a party-over-policy effect differs from a temporal framing effect in that both increases and decreases in policy support are expected, while previous research hypothesized that past framings exclusively increase conservatives' attitudes towards liberal political goals [Baldwin & Lammers, 2016; Lammers & Baldwin, 2018].

3.1. Participants

A sample of Swiss citizens (N = 503, 263 women) was recruited online via a market research institute. We aimed for the same sample size as in Experiment 1. All participants were required to have the Swiss nationality and to be eligible to vote. The mean age was 50.0 years (SD = 15.5). Distributions of sex, age, and political orientation closely represented the general Swiss population (see Table 2 in the Appendix). The completion of the experiment took about 5 min for which participants were compensated with 2€.

3.2. Procedure and experimental design

The procedure was identical to Experiment 1, with the exception that we varied messenger party affiliation (explicit identity cue) instead of temporal framing (implicit identity cue). Consequently, all participants read the future-framed version of the newspaper article. We systematically varied whether the author of the article, anonymously mentioned at the head of the article (see the Appendix for the complete stimuli), was member of a Swiss political party from the right (i.e., conservative): *Swiss People's Party* (n = 252) or the left (i.e., liberal): *Social Democratic Party of Switzerland* (n = 251) of the Swiss political spectrum. Fourty-one participants wrongly answered at least one of the attention questions and were replaced with newly recruited participants to maintain the planned sample size of 500. Cronbach's alpha again indicated high internal consistency of the endorsement measure (α = 0.86, 95%CI[0.84, 0.88]). Similar to the sample of Experiment 1, overall support of the climate policy was moderate (M = 4.04, SD = 1.44), as was overall endorsement (M = 3.10, SD = 0.87). Table 5 in the Appendix shows the bivariate correlations between all variables included in Experiment 2.[2]

3.3. Results and discussion

First, we analyzed the influence of messenger party affiliation on participants' perception of the political orientation of the messenger. Perceived political orientation of the messenger again leaned towards liberal (M = 4.6, SD = 1.9) but was largely normally distributed (skewness = 0.24, kurtosis = 2.52). The results of a linear regression

[2] At the end of Experiment 2, participants additionally reported to what extent they were surprised that a member of the conservative/liberal party advocated for the climate policy from 1 = *not surprised at all* to 6 = *very surprised*. In line with our expectations, participants were less surprised when the messenger was affiliated to the liberal (M = 2.67, SD = 1.48) than to the conservative party (M = 3.75, SD = 1.44; $t(501)$ = 12.66, $p < .001$).

confirmed that the messenger was perceived as more conservative when affiliating to a conservative political party than when being affiliated to a liberal political party (b-path in Fig. 1; $t(501) = 8.44$, $b = 1.31$, $se = 0.16$, $p < .001$). Although this result was unsurprising, the distribution of perceived political orientation showed considerable variation, supporting the importance of other influences such as temporal framing on individuals' judgments (see Fig. 4).

Next, we analyzed the data for the occurrence of a party-over-policy effect (e-path in Fig. 1). We computed separate linear regressions with participants' self-reported political orientation, messenger party affiliation, and the interaction of both as predictors of endorsement and policy support. For endorsement, an ANOVA revealed a statistically significant main effect of participants' political orientation, no main effect of messenger party affiliation, and the expected interaction of participants' political orientation and messenger party affiliation ($F(1, 499) = 11.45$, $p < .001$, $F(1, 499) = 0.20$, $p = .652$, and $F(1, 499) = 10.89$, $p = .001$, respectively). In line with our expectations, the regression coefficient of the interaction indicated that conservative party affiliation of the messenger attenuated the negative relationship between participants' political orientation and endorsement ($t(499) = 3.3$, $b = 0.15$, $se = 0.05$, $p = .001$). Importantly, however, a subsequent analysis of the Johnson-Neyman intervals of the interaction effect [Hayes & Matthes, 2009; Potthoff, 1964] revealed that conservative party affiliation not only increased the endorsement of more conservative participants (i.e., with political orientation > 6.77) but also decreased the endorsement of more liberal participants (i.e., with political orientation < 4.29; see the left plot of Fig. 5).[4]

For climate policy support a similar pattern emerged. An ANOVA revealed a main effect of participants' political orientation, no main effect of messenger party affiliation, and the expected interaction of participants' political orientation and messenger party affiliation ($F(1, 499) = 16.66$, $p < .001$, $F(1, 499) = 0.20$, $p = .66$, and $F(1, 499) = 4.91$, $p = .027$, respectively). An inspection of the interaction regression coefficient revealed an attenuation of the negative relationship between participants' political orientation and policy support when the messenger was affiliated to the conservative as opposed to the liberal political party ($t(499) = 2.22$, $b = 0.17$, $se = 0.08$, $p = .027$). However, the Johnson-Neyman intervals of the interaction indicated that conservative party affiliation only led to a decrease in policy support among more liberal participants (i.e., with political orientation < 3.28), whereas policy support of conservatives remained unchanged (i.e., upper limit of the JNI > 10; see the right plot of Fig. 5). This one-sided effect was unexpected, since conservatives tend to be more susceptible to explicit identity cues than liberals [e.g., Unsworth & Fielding, 2014].

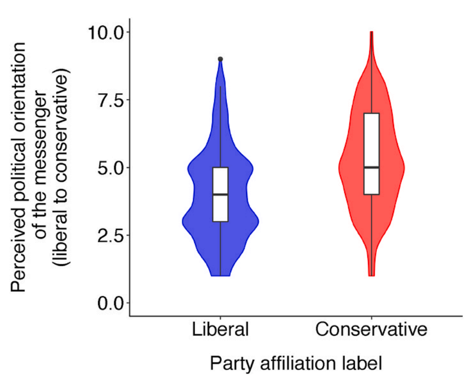

Fig. 4. Perception of messenger political orientation in Experiment 2. Box- and density plots illustrating the influence of explicit party affiliation on the perceived political orientation of the messenger. Liberal party affiliation resulted in the messenger being perceived as more liberal and conservative party affiliation resulted the messenger being perceived as more conservative. Perceived political orientation was reported on a scale from 1 = *extremely left* to 10 = *extremely right*. Points show one outlier that is situated more than 1.5 times the interquartile range away from the upper limit of the box plot.

To explore the underlying mechanisms of these findings, we computed two moderated mediation models testing whether perceived political orientation mediated the direct effect of explicit party affiliation on endorsement and policy support (lower triangle in Fig. 1). In line with the findings from Experiment 1, the results showed that perceived political orientation of the messenger, moderated by individuals' own political orientation, mediated the relationship between party affiliation and endorsement ($MM_{index} = 0.069$, $95\%CI[0.035, 0.106]$) as well as policy support ($MM_{index} = 0.096$, $95\%CI[0.041, 0.156]$). When accounting for the moderated mediation through the perception of the messenger, the interaction of party affiliation with individuals' own political orientation turned non-significant for endorsement ($F(1, 497) = 2.12$, $p = .15$) as well as policy support ($F(1, 497) = 0.62$, $p = .43$; see Table 1 in the Appendix for detailed results of the mediation analysis). As for Experiment 1, the mediation results should be interpreted with caution due to the potential influence of unobserved variables.

In summary, Experiment 2 illustrated that a less ambiguous explicit cue of political messenger identity similarly influenced individuals' perceived political orientation of the messenger (b-path in Fig. 1). In line with our expectations, the resulting change in the perceived political orientation was accompanied by changes in participants' endorsement and climate policy support as a function of their own political orientation (e-path). More specifically, in line with previous research [Cohen, 2003; Iyengar et al., 2019], liberals' *and* conservatives' endorsement and policy support tended to be the highest when the explicit party affiliation of the messenger aligned with their political orientation (Fig. 5). Moreover, in line with the results from Experiment 1, perceived political orientation was found to mediate the effect of party affiliation on endorsement and policy support, to the extent that the direct interaction between party affiliation label and political orientation disappeared when accounting for the mediator. Given this similarity in the underlying process found in Experiment 1 and 2, and in the light of liberals also being affected by the explicit political identity cue in Experiment 2, the question arises how liberals and conservatives are affected by a combination of temporal framing and party affiliation. Additionally, the presence of an explicit political identity cue may facilitate the integration of implicit identity cues, and thus the occurrence of a temporal framing effect [Diamond, 2020; Morris et al., 2008; Unsworth & Fielding, 2014]. To answer these questions and conclude our line of experiments, Experiment 3 was designed to investigate how temporal framing and political party affiliation jointly influence individuals' perception of political orientation, their endorsement and climate policy support.

4. Experiment 3

Experiment 3 tested whether the implicit and explicit political identity cues used in Experiment 1 and 2 independently contribute to individuals' perception of the political orientation of the messenger (Hypothesis 3). We expected that a past-framed message and a conservative party affiliation of the messenger both result in a perception of the messenger to be more conservative (a- and b-path in Fig. 1). Conversely, a future-framed message and a liberal party affiliation of the messenger were expected to result in a perception of the messenger to be more liberal. We moreover tested the influence of explicit and implicit identity cues on individuals' endorsement and climate policy support (c- and e-path).

4.1. Participants

A sample of Swiss citizens ($N = 1002$, 502 women) was recruited online via a market research institute. We aimed for the same sample sizes per experimental condition as in Experiment 1 and 2. All participants were required to have the Swiss nationality and to be eligible to vote. The mean age was 47.04 years ($SD = 26.7$). Distributions of sex,

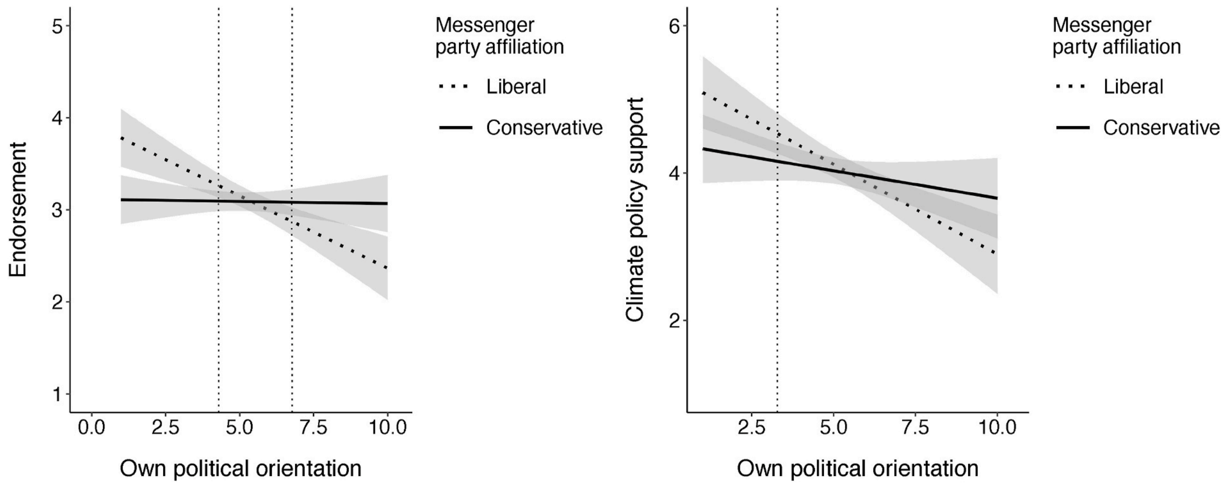

Fig. 5. Endorsement and climate policy support in Experiment 2. The interaction of messenger party affiliation with participants' own political orientation. While a conservative party affiliation of the messenger increased the endorsement of conservatives, it also decreased endorsement of liberals. For climate policy support, a conservative party affiliation only decreased the policy support of liberals, while there was no statistically significant increase in policy support among conservatives. Grey areas represent 95% confidence intervals around the regression lines. Dotted lines indicate the Johnson-Neyman intervals (JNI) for the interaction of messenger party affiliation and participants' political orientation, outside of which the difference between conservative and liberal party affiliation are statistically significant. For climate policy support the upper bound of the JNI is not displayed because it lies outside the possible values of political orientation (i.e., political orientation > 10).

age, and political orientation closely represented the general Swiss population (see Table 2 in the Appendix). The completion of the experiment took about 5 min for which participants were compensated with 2€.

4.2. Procedure and experimental design

In Experiment 3, we fully crossed the experimental designs of Experiment 1 and 2 while keeping all procedures constant. Accordingly, participants either read a future-framed article by a liberal party member ($n = 291$), a past-framed article by a liberal party member ($n = 222$), a future-framed article by a conservative party member ($n = 271$) or a past-framed article by a conservative party member ($n = 218$). To exclude the possibility that the perceived political orientation measure was influenced by the preceding subsidy support measure, half of the participants either first reported their voting intentions or their perception of the messenger's political orientation. Participants' perception of the political orientation of the messenger was not influenced by reporting order ($t(1000) = 0.76$, $b = 0.09$, $se = 0.12$, $p = .446$). Eighty-three participants wrongly answered at least one of the attention questions and were replaced with newly recruited participants to maintain the planned sample size of 1000. Cronbach's alpha again yielded a high internal consistency of the endorsement measure ($\alpha = 0.84$, $95\%CI[0.82, 0.86]$). Similar to the sample of Experiment 1 and 2, overall support of the climate policy was moderate ($M = 4.03$, $SD = 1.41$), as was endorsement ($M = 3.16$, $SD = 0.82$). Table 6 in the Appendix shows the bivariate correlations between all variables included in Experiment 3.[3]

4.3. Results and discussion

First, we analyzed the influence of temporal framing and messenger party affiliation on participants' perception of the political orientation of the messenger (a- and b-path in Fig. 1). Overall, perceived political orientation of the messenger was somewhat more liberal ($M = 4.9$, $SD = 1.9$) and again largely normally distributed (skewness $= 0.25$, kurtosis $= 2.87$). We computed a linear regression model including temporal framing, messenger party affiliation, and their interaction as predictors of the perceived political orientation of the messenger. The main effects of both temporal framing and messenger party affiliation were statistically significant, while their interaction was not ($F(1, 998) = 14.85$, $p < .001$, $F(1, 998) = 97.18$, $p < .001$, and $F(1, 998) = 0.00$, $p = .95$, respectively). In line with our predictions, the regression coefficients indicated that the messenger was perceived as more conservative in the past than the future framing condition (Hypothesis 1), and when affiliating to a conservative as compared to a liberal political party ($t(998) = 2.65$, $b = 0.42$, $se = 0.16$, $p = .008$, and $t(998) = 6.58$, $b = 1.11$, $se = 0.17$, $p < .001$). Temporal framing and messenger party affiliation thus seem to independently influence the perception of the political

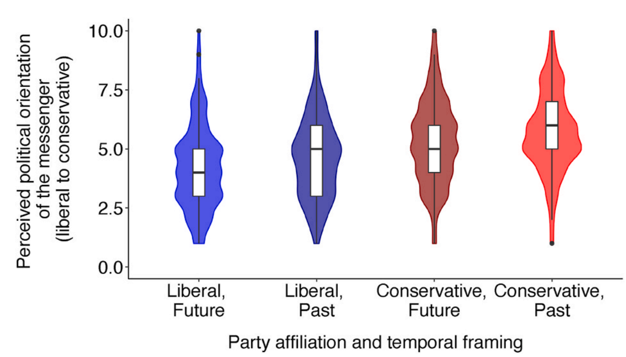

Fig. 6. Perception of messenger political orientation in Experiment 3. Box- and density plots illustrating the influence of the crossed temporal framing and messenger party affiliation manipulations on the perceived political orientation of the messenger. Liberal party affiliation as well as the future framing resulted in a more liberal perception of the political orientation of messenger, while conservative party affiliation and the past framing resulted in a more conservative perception of the political orientation of the messenger. Perceived political orientation was reported on a scale from 1 = *extremely left* to 10 = *extremely right*. Points show outliers that are situated more than 1.5 times the interquartile range away from the upper or lower limit of the box plot.

[3] As in Experiment 2, we explored to what extent participants were surprised that a member of the conservative/liberal party advocated for the climate policy, as a function of the temporal framing used. Participants were again less surprised when the messenger affiliated to the liberal ($M = 2.76$, $SD = 1.43$) than to the conservative party ($M = 4.09$, $SD = 1.46$; $t(1000) = 14.6$, $p < .001$), while temporal framing did not have an effect ($t < 1$).

orientation of the messenger (independence of a- and b-path; see Fig. 6).

Next, we analyzed the data for the occurrences of a temporal framing and party-over-policy effect (c- and e-path). We computed separate linear regressions with participants' self-reported political orientation, temporal framing, messenger party affiliation, and the interactions of both manipulations with participants' political orientation as predictors of endorsement and policy support.

For endorsement, the results of an ANOVA revealed no main effects of temporal framing, messenger party affiliation, and no interaction of both manipulations ($F(1, 994) = 0.52, p = .47, F(1, 994) = 0.64, p = .42$, and $F(1, 994) = 0.00$, $p = .97$). However, both interactions of participants' political orientation with the temporal framing and with messenger party affiliation condition were statistically significant ($F(1, 994) = 8.49, p = .004$, and $F(1, 994) = 15.88, p < .001$). The three-way interaction of both manipulations with participants' political orientation was not significant ($F(1, 994) = 0.67, p = .41$). A post-hoc analysis of the simple interaction coefficients corroborated a party-over-policy effect, in line with the results of Experiment 2 ($t(994) = 3.23$, $b = 0.15, se = 0.05, p = .001$). The analysis of the Johnson-Neyman intervals [Hayes & Matthes, 2009; Potthoff, 1964] showed that conservative party affiliation increased endorsement of conservatives (i.e., with political orientation > 6.62), but also decreased endorsement of liberals (i.e., with political orientation < 4.80; see dotted lines in the left plot of Fig. 7). Additionally, the post-hoc analysis also corroborated a temporal framing effect ($t(994) = 2.46$, $b = 0.11$, $se = 0.05$, $p = .014$). The Johnson-Neyman intervals indicated that past framing not only increased the endorsement of conservative participants (i.e., with political orientation > 6.26), but also decreased the endorsement of liberal participants (i.e., with political orientation < 2.98; see dashed lines in the left plot of Fig. 7).[4]

For climate policy support, the results of an ANOVA revealed again no main effects of neither temporal framing nor messenger party affiliation, and no interaction of the two ($F(1, 994) = 0.79, p = .37, F(1, 994) = 1.25$, $p = .26$, and $F(1, 994) = 0.12$, $p = .73$). However, the results yielded an interaction of temporal framing with participants' political orientation ($F(1, 994) = 3.85, p = .050$), while there was no statistically significant interaction of messenger party affiliation with participants' political orientation on policy support ($F(1, 994) = 1.66, p = .198$). The three-way interaction of both manipulations with participants' political orientation was again not statistically significant ($F(1, 994) = 1.18, p = .277$).

A post-hoc analysis of the significant interaction between the temporal framing manipulation and participants' political orientation revealed that the past framing attenuated the negative relationship between political orientation and climate policy support ($t(994) = 2.09, b = 0.16, se = 0.08, p = .037$). In line with the original temporal framing effect [Baldwin & Lammers, 2016], subsequent analyses of the Johnson-Neyman intervals indicated that the past framing exclusively increased the climate policy support of conservative participants (i.e., with political orientation > 7.01, see dashed line in the right plot of Fig. 7), but not of liberal participants (i.e., lower limit of the JNI < 1). The non-significant interaction between messenger party affiliation and participants' political orientation did not allow to compute Johnson-Neyman intervals.

To further explore whether the coherence of a past-framed message being communicated by a conservative messenger led to particularly high climate policy support by conservatives, we separately introduced all four experimental conditions (i.e., liberal-future, liberal-past, conservative-future, conservative-past) into a linear regression. The results indicated that merely in the conservative-past condition the relationship between political orientation and climate policy support was non-significant ($t(994) = -1.39$, $b = -0.08$, $se = 0.06$, $p = .17$), while the coefficients of the other experimental conditions continued to indicate a negative relationship (all $ps < .001$). This suggests that past framing may be particularly effective in increasing conservatives' climate policy support when being communicated by a conservative party member.

Finally, to explore the underlying process, we analyzed whether perceived political orientation mediates the effect of both manipulations on endorsement and climate policy support. In line with the findings from Experiment 1, mediation analyses indicated that perceived political orientation of the messenger mediated the effect of temporal framing on endorsement ($MM_{index} = 0.029$, $95\%CI[0.013, 0.049]$) and policy support ($MM_{index} = 0.03$, $95\%CI[0.011, 0.054]$). When accounting for the indirect effect, the interaction of temporal framing with individuals' political orientation turned non-significant for both endorsement and policy support ($F(1, 996) = 2.84$, $p = .093$ and $F(1, 996) = 1.67$, $p = .196$, respectively). Similarly, and in line with the findings from Experiment 2, perceived political orientation again mediated the effect of explicit party affiliation on both endorsement ($MM_{index} = 0.071$, $95\%CI[0.048, 0.098]$) and policy support ($MM_{index} = 0.077$, $95\%CI[0.043, 0.116]$). When accounting for the indirect effect, the interaction of messenger party affiliation and individuals' political orientation continued to be statistically significant for endorsement ($F(1, 996) = 6.49, p = .011$) but not for policy support ($F(1, 996) = 0.32, p = .57$; see Table 1 in the Appendix for detailed results of the mediation analysis).

In summary, Experiment 3 extends our findings on temporal message framing and messenger party affiliation from Experiment 1 and 2 by showing that both identity cues exert unique influence on participants' perception of the messenger (a- and b-path in Fig. 1). The analysis of participants' endorsement moreover replicated the party-over-policy effect observed in Experiment 2 (e-path), and in contrast to Experiment 1 also replicated the original temporal framing effect [c-path; Baldwin & Lammers, 2016; Lammers & Baldwin, 2018]. As in Experiment 1 and 2, moderated mediation analyses suggested a mediating role of perceived political orientation of the messenger for both, the temporal framing effect and the party-over-policy effect. Moreover, the presence of the explicit political identity cue in Experiment 3 may have facilitated the integration of the implicit identity cues contained in the temporal framing by activating individuals' political identity [Zaller, 1992; Diamond, 2020; Morris et al., 2008; Unsworth & Fielding, 2014]. Previous research has similarly observed a temporal framing effect in the presence of an explicit political identity cue [Baldwin & Lammers, 2016], although not consistently [Kim et al., 2021].

Moreover, a past framing did not only increase the endorsement of conservatives, but also decreased the endorsement of liberals. This finding is in line with previous research finding a negative effect of past framings on liberals' environmental outcomes [Kim et al., 2021; Stanley et al., 2021], and provides supporting evidence for an indirect framing effect suggested by our integrated theoretical account. However, lower endorsement by liberals may be of limited practical relevance, since the present findings converged with the original research by only showing an increase of conservatives' policy support in the past framing condition whereas liberals' policy support remained unaffected [Baldwin & Lammers, 2016]. Importantly, merely the combination of a past framing with a conservative party affiliation led to a complete attenuation of the negative relationship between political conservatism and climate policy support, suggesting that a certain threshold of the messenger being perceived as conservative may be necessary to increase conservatives' climate policy support.

[4] Separate linear regressions with each of the three endorsement measures as outcomes showed that the interaction between party affiliation label and individuals' political orientation was statistically significant for messenger liking and messenger voting intention, but not for message agreement which more strongly focused on the content of the message rather than on its messenger, see Table 3 in the Appendix for more details. We thank one anonymous reviewer for suggesting this more fine-grained analysis of the endorsement outcome.

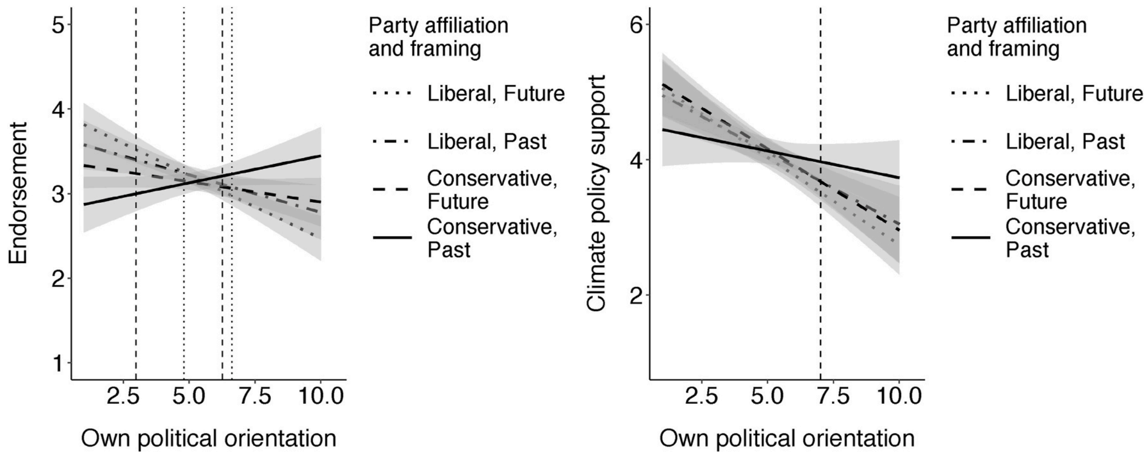

Fig. 7. Endorsement and climate policy support in Experiment 3. The left plot shows how the experimental manipulations interact with participants' self-reported political orientation to shape their endorsement. While a future framing by a liberal party member resulted in the highest endorsement by more liberal participants, it led to the lowest endorsement by conservative participants. Conversely, a past framing by a conservative party member resulted in the highest endorsement by more conservative participants, while it led to the lowest endorsement by liberal participants. The right plot shows how the experimental manipulations interact with participants' political orientation to shape their climate policy support. Past framing and conservative party affiliation independently increased climate policy support by conservatives. The combination of past framing and conservative party affiliation attenuated the effect of political conservatism on climate policy support and did not decrease policy support by liberals. Grey areas represent 95% confidence intervals around the regression lines. Dotted lines indicate the Johnson-Neyman intervals (JNI) for the interaction of messenger party affiliation and participants' political orientation, outside of which the difference between conservative and liberal party affiliation are statistically significant. Dashed lines indicate the JNI for the interaction of temporal framing and participants' political orientation, outside of which the difference between past and future framing are statistically significant. For climate policy support the lower bound of the JNI is not displayed because it lies outside the possible values of political orientation (i.e., political orientation < 10).

5. General discussion

Does past-framed messaging constitute a simple solution to narrow the political divide on environmental issues? Our findings show a more complex picture while at the same time providing some evidence supporting the original findings of a temporal framing effect [Baldwin & Lammers, 2016]. In the present research we argue that temporal message framings convey implicit cues of a messenger's political orientation, with past framings indicating a more conservative messenger and future framings indicating a more liberal messenger. This perspective allowed us to investigate to what extent implicit temporal identity cues operate similarly to more explicit cues of messengers' political identity such as the affiliation to a political party [Cohen, 2003; Iyengar et al., 2019]. Our findings moreover offer an explanation for previous unsuccessful replications of the temporal framing effect, finding that both conservatives and liberals may be similarly affected by temporal framings [Stanley et al., 2021; Kim et al., 2021].

Across three experiments, our results showed that a past-framed climate policy and the explicitly stated affiliation to a conservative political party resulted in the messenger being perceived as more conservative, suggesting some similarity between implicit and explicit political identity cues. In support of this integrative perspective, mediation analyses suggested that changes in messenger perception underlie both the effects of temporal framing and party affiliation on individuals' judgments and climate policy support.

However, altered messenger perception was not always accompanied by partisans expressing more favorable attitudes or policy support. While in Experiment 1 temporal framing did not alter neither participants' endorsement nor climate policy support, in Experiment 3 the identical temporal framing altered both outcomes as a function of partisans' own political orientation. Since the only difference between both experiments was that Experiment 3 manipulated the party affiliation of the messenger (i.e., liberal vs. conservative) in addition to the temporal framing, the presence of this explicit political identity cue seems to have played a facilitating role for observing a temporal framing effect. Some previous findings of a temporal framing effect similarly manipulated the explicit political identity of the messenger in parallel to a temporal framing manipulation [Baldwin & Lammers, 2016], providing a potential explanation for the null-effect observed in Experiment 1 and other failures to replicate the effect [Stanley et al., 2021].

When simultaneously manipulating temporal framing and messenger party affiliation in Experiment 3, we found that both manipulations influenced partisans' endorsement to a comparable extent. A past-framed message and conservative party affiliation of the messenger increased the endorsement by conservatives while decreasing the endorsement by liberals. This effect points to a similarity of implicit and explicit political identity cues, offering an explanation for previous findings of liberals being equally affected by temporal framings as conservatives [Stanley et al., 2021; Kim et al., 2021]. Interestingly, the results for climate policy support in Experiment 3 did not follow the same pattern. In line with the original account and in support of the practical relevance of temporal framings, we found a past-framing to increase conservatives' policy support while leaving liberals' climate policy support unchanged [Baldwin & Lammers, 2016]. Moreover, only a past-framing communicated by a conservative party member entirely attenuated the negative relationship between political conservatism and climate policy support. Given that this combination resulted in the most conservative perception of the messenger and that changes in perception seem to indirectly explain part of the observed direct effects, it could be that the perception of the messenger has to pass a certain threshold of conservatism in order for conservatives to increase their policy support.

5.1. Theoretical implications

The present research contributes to the understanding of how individuals form their perception of the political identity of a messenger based on implicit and explicit identity cues, and the influence of both types of cues on partisans' judgments and environmental outcomes [Baldwin & Lammers, 2016; Lammers & Baldwin, 2018; Kim et al., 2021; Iyengar et al., 2019; Van Boven et al., 2018]. We hypothesized that temporal framings serve as implicit cues of the political identity of a messenger, a perspective that integrates research on framing effects with the extensive literature on partisan effects [Iyengar et al., 2019; Van Boven et al., 2018; Cohen, 2003].

Our research addressed the question whether implicit and explicit cues of political identity contribute to individuals' perception of the political orientation of a messenger equally and independently, or if one effect dominates over the other. In line with theory on cue utilization in judgment formation [Brunswik, 1956; Anderson, 1962; Slovic, 1966], we found that individuals make use of implicit identity cues contained in temporal framings as well as explicit cues such as party affiliation to form their judgments of the political orientation of the messenger. Moreover, in the presence of both implicit and explicit political identity cues, both cues jointly and independently shaped individuals' perception of the political orientation of the messenger. This supports evidence that individuals integrate small numbers of cues relating to the same unobserved criterion in an additive manner [Anderson, 1962; Slovic, 1966; Hurst & Stern, 2020; Voelkel & Willer, 2019; Kleissner & Jahn, 2021; Derous & Decoster, 2017].

Our results moreover support the idea that perceived political orientation should be considered as a fine-grained continuum influenced by multiple cues, accounting for nuanced differences between its two extremes "liberal" and "conservative" [Fielding et al., 2020; Knippenberg, 1999]. This raises the questions whether other implicit identity cues, as for example contained in moral framings [Feinberg & Willer, 2013; Wolsko et al., 2016], would additionally influence the perception of messenger identity. While some research has suggested that explicit political identity cues may eliminate the influence of moral framings because they do not provide any new information about the identity of a messenger [Fielding et al., 2020; Feinberg & Willer, 2019], other research has found conservatives' support for a Democratic presidential candidate and environmental policies to be independently influenced by moral framings and explicit political identity cues [Voelkel & Willer, 2019; Hurst & Stern, 2020], similar to the present findings for temporal framings. It remains an open question whether different implicit identity cues (e.g., moral and temporal framings) independently shape the perception of the political identity of a messenger.

Our replication of a temporal framing effect in Experiment 3 but not in Experiment 1 may point to a potential boundary condition of the effect. Since the only difference of Experiment 3 was that the party affiliation of the messenger was revealed, the presence of this explicit political identity cue seems to have influenced the occurrence of a temporal framing effect. One explanation for this finding may be that explicitly labeling political identity facilitated the integration of the implicit temporal political identity cues into individuals' judgments by activating their political identity [Zaller, 1992; Unsworth & Fielding, 2014; Morris et al., 2008]. Only a clear signal of the context being political may trigger motivated reasoning mechanisms to form judgments conforming with the position of one's political group [Kahan et al., 2012; Druckman & McGrath, 2019]. This idea is supported by related research that did not find an effect of moral framing when communicated by a non-partisan source [Hurst & Stern, 2020], potentially because a non-partisan source did not activate individuals' political identity. While the absence of an explicit political identity cue may explain other failures to replicate a temporal framing effect [Stanley et al., 2021], some research did not replicate the effect despite the presence of an explicit political identity cue [Kim et al., 2021]. Future research should investigate the role of identity-activating contexts on the temporal framing effect within one experimental design, since the present work only provided evidence across two separate experiments.

5.2. Practical implications

Our findings have relevant practical implications for the advocacy of climate policies. In line with the original temporal framing account [Baldwin & Lammers, 2016; Lammers & Baldwin, 2018], increasing the use of past-oriented language when promoting climate policies may, in some circumstances, increase climate policy support of more conservative citizens.

Our results showed that a past framing increased climate policy support of conservatives but did not decrease the policy support of liberals. This suggests that there is a qualitative difference between liberals' endorsement of a political messenger and a message and their ultimate support of a climate policy. While liberals may show lower endorsement of a past-framed message because they perceive its messenger to be conservative, cues of opposing political identity do not seem to diminish their approval of climate policies [Unsworth & Fielding, 2014; Doell et al., 2021; Kahneman & Tversky, 1984].

However, our findings of Experiments 3 showed that the negative relationship between political conservatism and climate policy support was merely attenuated when a past framing was communicated by a messenger affiliating to a conservative political party. This finding extends existing evidence suggesting that conservatives more strongly rely on messenger party affiliation for their judgments and decisions than liberals [Unsworth & Fielding, 2014; Morris et al., 2008; Linde, 2018]. Future research may investigate whether reinforcing factors of conservatives' partisan effects in response to explicit political identity cues, such as higher education [Kahan et al., 2012], also reinforce temporal framing effects.

A boundary condition of the temporal framing effect suggested by the present research is that it needs to be embedded into a context where individuals' political identity is activated. From a practical perspective, this condition may often be given - in many occasions political identity is made salient since political goals tend to be advocated for by political parties.

However, the benefit of activating the political identity of individuals in order to facilitate the occurrence of a temporal framing effect should be weighed against potential undesirable effects [Doell et al., 2021]. Research has for example shown that activating political identity even in political consensus situations polarizes partisans [Linde, 2018]. This may contribute to a further exacerbation of the political and societal divide. Similarly, our findings on individuals' endorsement support a cautious use of identity-activating cues and temporal framing: message and messenger endorsement strongly polarized between liberals and conservatives when the political party affiliation of the messenger was communicated. Climate change advocacy that entirely avoids to provoke this type of out-group opposition may altogether reduce the political divide on climate change [Hornsey et al., 2018]. For example, focusing citizens on the actual merits of a given policy may increase policy preference independent of partisan sentiments [Wong-Parodi & Fischhoff, 2015]. Similarly, activating identities that are shared between liberals and conservatives and are congruent with a motivation to mitigate climate change, such as the identity as a parent or grandparent [Diamond, 2020; Doell et al., 2021], may also avoid a further politicisation of environmental issues [Unsworth & Fielding, 2014; Bain et al., 2012].

5.3. Limitations and future research

The present research has some limitations. While we observed striking differences in the observation of a temporal framing effect between Experiment 1 and 3, which exclusively differed in the orthogonal manipulation of messenger party affiliation, we did not directly test the influence of the presence (vs. absence) of an explicit political identity cue. This limits the conclusions that can be drawn from this comparison, and should be followed up by research more closely investigating and measuring the potential identity-activating effect of explicit political identity cues and their relevance for the integration of more subtle implicit identity cues into individuals' judgments and decisions. It would moreover be interesting to investigate whether different explicit cues of political identity, such as political party affiliation, self-identification, or membership to (environmental) associations have similarly identity-activating effects.

Another limitation is that our findings may not transfer to countries with a political landscape that substantially differs from the one in Switzerland. We measured political orientation on a scale from *left* to *right*, which is the common measure of political orientation in a European context [e.g., Hahnel et al., 2020], but which is not conceptually equivalent to political orientation measured on a scale from *liberal* to *conservative*, typically used in the U.S. [Hornsey et al., 2018]. While previous research has found convergent findings using both measures across the U.S., the U.K., and Germany [Lammers & Baldwin, 2018], it remains an open question whether comparable results would be obtained in non-Western countries where conservatism and right-wing political orientation are less strongly associated [Jost et al., 2009]. Based on the theoretical account advanced here, a temporal framing effect should occur as long as past/future-oriented language is a valid cue for the political identity of a messenger.

Moreover, the present research merely investigated climate policy support in the context of electric vehicle subsidies. Subsidizing electric vehicles is only one of many climate policies and exclusively focusing on this topic may reduce the generalizability of our findings to other types of climate policies. Previous research has found partisan cues to exert a significant influence on the support of a wide range of climate policies [Linde, 2018], however with varying influence, showing for example a weaker effect for subsidies of green electricity and a stronger effect for a CO_2 tax [Linde, 2018]. Future research should extend the present work by investigating temporal framing effects across a wider range of climate policies.

Additionally, it would be valuable to extend the investigation of temporal framing effects to traditionally more conservative policy domains. Based on the theoretical account presented here, a future (vs. past) framing may increase liberals' support of conservative policies (e.g., increasing military investments) by resulting in a more liberal perception of the messenger. On the other hand, the present findings and other research suggests that liberals' policy support may be generally less susceptible to political identity cues [Unsworth & Fielding, 2014], leading to the prediction that a future framing may increase liberals' endorsement but not their support of a clearly conservative policy. Future research should provide additional evidence on liberals' policy support seeming to be less strongly affected by temporal framings than conservatives' policy support.

Finally, our data allowed merely limited conclusions about the role of messenger perception in explaining the effects of temporal framing and party labeling, due to the potential influence of unobserved variables [Rohrer et al., 2022; Bullock & Green, 2021]. Methodological approaches relaxing the assumption of mediator and outcome being uncorrelated outside of the experiment [e.g., implicit-mediation designs; Bullock & Green, 2021], could provide a viable option to assess the causal influence of perceived political orientation and other potential mediators [e.g., nostalgia; Lammers & Baldwin, 2018] for the framing and identity labeling effects investigated here.

5.4. Conclusion

Previous research has suggested that past framings may be used by a wide range of actors to increase conservatives' climate policy support [Lammers & Baldwin, 2018; Baldwin & Lammers, 2016]. The present findings partly support this claim by showing a positive effect of temporal framing on conservatives' climate policy support when its messenger explicitly affiliates to a political party - independent of the party being liberal or conservative. However, this does not mean that explicit political identity cues do not influence partisans' policy support: We found that only a combination of past framing and conservative party affiliation, which was associated with the messenger being perceived as most conservative, entirely attenuated the negative relationship of political conservatism and climate policy support. The most promising solution to win conservatives for climate policy support through past framing thus seems to be its delivery by a conservative party member. Closing the political divide on climate policies appears to remain a question of winning the support of conservative elites and not merely a question of the right framing.

Data and code availability

All data and code used to generate results and figures of the present work are publicly available at https://doi.org/10.17605/OSF.IO/YFXMC.

Ethics statement

This research was approved by the Ethics commission of the Faculty for Psychology and the Educational Sciences of the University of Geneva, and informed consent was received from all participants.

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CRediT authorship contribution statement

Mario Herberz: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration. **Tobias Brosch:** Conceptualization, Methodology, Writing – review & editing, Supervision, Funding acquisition. **Ulf J.J. Hahnel:** Conceptualization, Methodology, Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

None.

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Appendix

Stimuli used for the future framing condition

Translated from German. For the original texts, please contact the authors.

Electromobility: From the Present into the Future

[Liberal party condition] Author: *anonymised within the context of this research project*, member of the Swiss Social Democratic Party (SP).

[Conservative party condition] Author: *anonymised within the context of this research project*, member of the Swiss People's Party (SVP).The future of electromobility will show that the electric car will play a central role in our social development. It will be celebrated by future generations as an important technical achievement and will be part of a major economic boom whose success story will not depend on the availability of cheap oil for carburettor fuels.

Electric propulsion will not only be a promising advance for the science and economy of the future, but will also change people's daily lives. Today's infrastructure allows electric cars to be conveniently charged, and running on domestically produced electricity will enable independence from foreign energy sources in the future.

Through the development of the electric car, we can learn many things today that will continue to be of benefit to society in the future. In particular, we will benefit in the future from the many new forms of mobility that will be central components of our lives.

Today's upsurge in electromobility represents an unprecedented opportunity, and we now bear the responsibility for its success. I therefore advocate that additional public subsidies for electromobility should now be decided in a referendum. I feel it is our duty as Swiss to drive forward social visions of the future and to shape our mobility today in a way that is in line with progress.

Stimuli used for the past framing condition

Translated from German. For the original texts, please contact the authors.

Electromobility: From the Past to the Present

[Liberal party condition] Author: *anonymised within the context of this research project*, member of the Swiss Social Democratic Party (SP).

[Conservative party condition] Author: *anonymised within the context of this research project*, member of the Swiss People's Party (SVP).

The history of electromobility shows that as early as the 19th century, the electric car played a central role in social development. It was celebrated by our forefathers as an important technical achievement and was part of a significant economic boom before the availability of cheap oil for carburetor fuels interrupted this success story.

Electric propulsion not only represented a promising advance for 19th century science and business, but also promised to change people's daily lives. The infrastructure of the time allowed electric cars to be conveniently charged, and running on domestically produced electricity enabled potential independence from foreign energy sources.

Through the development of the electric car, our forefathers were able to learn many things that are still of use to society today. In particular, we benefit today from the many forms of mobility that are the legacy of previous generations, and which have become central parts of our lives.

Today's upsurge in electric mobility represents a resurgence of the achievements of previous generations, and we now bear the responsibility for its success. Therefore, I advocate that additional public subsidies for electromobility should now be decided in a referendum. I feel it is our duty as Swiss people to build on the ideas of our forefathers and to shape our mobility today in the way it was intended long ago.

Table 1
Summary of direct and indirect effects of temporal framing and party affiliation labels

Experiment	Dependent variable	c-path (moderated)	Moderated mediation index	a-path	b-path (moderated)	c'-path (moderated)
Exp 1 (temporal framing)	Endorsement	0.01(0.4)	0.03(0.01)[a]	0.35(0.15)*	0.07(0.01)***	−0.05(0.04)
	Policy support	0.03(0.07)	0.03(0.02)[a]	0.35(0.15)*	0.09(0.02)***	−0.05(0.07)
Exp 2 (party affiliation)	Endorsement	0.15(0.05)**	0.07(0.02)[a]	1.31(0.16)***	0.05(0.01)***	0.07(0.05)
	Policy support	0.17(0.08)*	0.10(0.03)[a]	1.31(0.16)***	0.07(0.02)***	0.06(0.08)
Exp 3 (temporal framing)	Endorsement	0.09(0.03)**	0.03(0.01)[a]	0.43(0.12)***	0.07(0.01)***	0.05(0.03)+
	Policy support	0.11(0.05)*	0.03(0.01)[a]	0.43(0.12)***	0.07(0.01)***	0.07(0.05)
Exp 3 (party affiliation)	Endorsement	0.13(0.03)***	0.07(0.01)[a]	1.11(0.11)***	0.06(0.01)***	0.08(0.03)*
	Policy support	0.07(0.05)	0.08(0.02)[a]	1.11(0.11)***	0.07(0.01)***	0.03(0.05)

Note. Linear regression coefficients with standard errors in parentheses. The column *c-path (moderated)* summarizes the direct effects of the interaction between experimental condition and individuals' political orientation on endorsement and policy support (c- and e-paths in Fig. 1). The remaining columns summarize the indirect effects obtained with PROCESS model 15 for moderated mediation (Hayes, 2017). The *a-path* column shows the effect of experimental condition on the proposed mediator: perceived political orientation of the messenger (a- and b-paths in Fig. 1). The *b-path* column shows the effect of the mediator, moderated by individuals' own political orientation, on the outcomes (d-path in Fig. 1). Finally, the *c'-path* column shows the c-path relationship when accounting for the mediating effect of messenger perception. ***$p < .001$, **$p < .01$, *$p < .05$, $^{+}p < .10$, [a] significant Bootstrap result.

Table 2
Comparison of samples with Swiss general population.

	Experiment 1	Experiment 2	Experiment 3	Swiss population
% female	51.3	52.4	50.1	50.4[b]
Age*	49.2 (14.0)	50.0 (15.5)	46.1 (15.2)	42.7[c]
Political orientation[a,*]	5.3 (1.8)	5.2 (1.7)	5.3 (1.7)	5.3 (2.0)[d]
N	507	503	1002	

Note. Descriptive statistics of the demographic variables sex, age, and political orientation across all experiments included in the present research and in comparison the general Swiss population.

[a] Measured on a scale from 1: "left" to 10: "right".

[b] [SFSO, 2022a].

[c] [SFSO, 2022b].

[d] Data retrieved from the World Values Survey Wave 7: 2017–2022, $N = 3042$; [Inglehart et al., 2022].

* Mean values with standard deviation, where available, in brackets.

Table 3
Temporal framing and party affiliation effects as a function of endorsement outcome

Endorsement outcome	Experiment 1 (temporal framing)	Experiment 2 (party affiliation)	Experiment 3 (temporal framing)	Experiment 3 (party affiliation)
Message agreement	0.007(0.05)	0.068(0.05)	0.096(0.04)**	0.058(0.04)
Messenger liking	0.011(0.05)	0.142(0.05)**	0.081(0.04)*	0.119(0.04)***
Messenger voting	0.013(0.05)	0.248(0.06)***	0.092(0.04)*	0.206(0.04)***
ANOVA for differences between endorsement outcomes	$F(2, 1006) = 0.01$	$F(2, 998) = 9.18$***	$F(2, 1996) = 0.12$	$F(2, 1996) = 11.79$***

Note. Separate ANOVAs were computed to test the three-way interaction between experimental condition, individuals' political orientation, and endorsement outcome in predicting endorsement. The results showed that for the interaction between party affiliation labels and individuals' political orientation in Experiment 2 and 3 individuals' endorsement differed as a function of endorsement outcomes. Subsequent linear regressions revealed that there was an interaction of party affiliation label and individuals political orientation for the messenger liking and the messenger voting outcome, but not for the message agreement outcome. In contrast, the interaction between temporal framing and individuals' political orientation in Experiment 1 and 3 did not differ as a function of endorsement outcome. ***$p < .001$,**$p < .01$, *$p < .05$.

Table 4
Raw correlations of variables included in Experiment 1.

	Age	Political orientation	Perceived political orientation	Message agreement	Author liking	Author voting	Endorsement index	Policy support
Sex	0.17***	.10*	−0.02	0.09*	0.07	0.07	0.08^{+}	0.05
Age		0.08^{+}	0.12**	−0.05	−0.04	−0.08^{+}	−0.06	−0.04
Political orientation			0.04	−0.19***	−0.16***	−0.18***	−0.20***	−0.24***
Perceived political orientation				0.11*	0.11*	0.12**	0.13**	0.13**
Message agreement					0.73***	0.71***	0.90***	0.72***
Author liking						0.74***	0.91***	0.63***
Author voting							0.91***	0.68***
Endorsement index								0.75***

Note. Correlation coefficients and their corresponding statistical tests correspond to Pearson's *r* for continuous variable pairs and Spearman's *rho* for variable pairs with at least one non-continuous variable. *Endorsement index* is the arithmetic mean of *Message agreement*, *Author liking*, and *Author voting* (see Methods). ***$p < .001$, **$p < .01$, *$p < .05$, $^{+}p < .10$.

Table 5
Raw correlations of variables included in Experiment 2.

	Age	Political orientation	Perceived political orientation	Surprise	Message agreement	Author liking	Author voting	Endorsement index	Policy support
Sex	0.24***	.10*	−0.05	0.03	−0.04	0.01	−0.09^{+}	−0.05	−0.02
Age		0.09^{+}	−0.07^{+}	0.08^{+}	−0.12**	−0.10*	−0.12**	−0.13**	−0.07
Political orientation			0.05	0.04	−0.13**	−0.17***	−0.10*	−0.15**	−0.18***
Perceived political orientation				−0.27***	0.16***	0.16***	0.16***	0.18***	0.21***
Surprise					−0.13**	−0.12**	−0.04	−0.11*	−0.16***
Message agreement						0.70***	0.63***	0.87***	0.72***
Author liking							0.71***	0.90***	0.59***
Author voting								0.89***	0.57***
Endorsement index									0.70***

Note. Correlation coefficients and their corresponding statistical tests correspond to Pearson's *r* for continuous variable pairs and Spearman's *rho* for variable pairs with at least one non-continuous variable. *Endorsement index* is the arithmetic mean of *Message agreement*, *Author liking*, and *Author voting* (see Methods). *Surprise* of participants that a member of the conservative/liberal party advocated for the climate policy was not included in our main analyses (see Footnotes 1 and 2). ***$p < .001$,**$p < .01$, *$p < .05$, $^{+}p < .10$.

Table 6
Raw correlations of variables included in Experiment 3.

	Age	Political orientation	Perceived political orientation	Surprise	Message agreement	Author liking	Author voting	Endorsement index	Policy support
Sex	0.04	.11***	−0.01	0.03	0.07*	0.01	−0.01	0.03	0.01
Age		0.07*	0.06*	0.02	−0.04	−0.03	−0.05^{+}	−0.05	−0.04
Political orientation			0.12***	0.01	−0.15***	−0.08**	−0.12***	−0.13***	−0.24***
Perceived political orientation				−0.22***	0.16***	0.06*	0.12***	0.13***	0.16***
Surprise					−0.10**	−0.08*	−0.06^{+}	−0.09**	−0.15***
Message agreement						0.63***	0.58***	0.84***	0.68***
Author liking							0.70***	0.89***	0.53***
Author voting								0.88***	0.54***
Endorsement index									0.67***

Note. Correlation coefficients and their corresponding statistical tests correspond to Pearson's r for continuous variable pairs and Spearman's *rho* for variable pairs with at least one non-continuous variable. *Endorsement index* is the arithmetic mean of *Message agreement*, *Author liking*, and *Author voting* (see Methods). *Surprise* of participants that a member of the conservative/liberal party advocated for the climate policy was not included in our main analyses (see Footnotes 1 and 2). ***$p <$ .001,**$p < .01$, *$p < .05$, $^{+}p < .10$.

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